

Report  
Of

**WEB DEVELOPMENT INTERNSHIP:  
BUILDING RESPONSIVE, INTERACTIVE AND DEPLOYED WEB  
APPLICATIONS USING HTML5, CSS3 AND JAVASCRIPT**

Submitted  
To

**School of Computer Science and Engineering  
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of  
**B.Tech CSE**



By

Roll Number of Student: 25SCS1003004257

**Name of Student: SUBHASH CHANDRA YADAV**

Batch: 2025–29

## CANDIDATE'S DECLARATION

I, SUBHASH CHANDRA YADAV, Roll No. 25SCS1003004257, do hereby declare that the internship report titled “Web Development Internship: Building Responsive, Interactive and Deployed Web Applications Using HTML5, CSS3 and JavaScript” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

A handwritten signature in blue ink, reading "Subhash Chandra". The signature is written in a cursive, flowing style.

Signature

Student Name: SUBHASH CHANDRA YADAV

Roll No.: 25SCS1003004257

## ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I would like to thank Thiranex – Skill Development & Future Tech for providing me with the opportunity to undergo this Web Development Internship, and for the structured tasks, mentorship and resources that helped me build the projects discussed in this report. I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous support and encouragement throughout this internship.

I express my gratitude to my parents for their blessings.

A handwritten signature in blue ink, reading "Subhash Chandra". The signature is written in a cursive, flowing style.

Signature

Student Name: SUBHASH CHANDRA YADAV

Roll No.: 25SCS1003004257

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## Internship Completion Certificate

The internship offer letter and the certificate of completion issued by Thiranex – Skill Development & Future Tech on successful completion of the Web Development Internship are attached below.

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SKILL DEVELOPMENT & FUTURE TECH

Date: 30 July 2026

TO,  
**Subhash Chandra Yadav**  
Intern ID: THX-JUL3026-504

Subject: Offer Letter for Internship in Web Development

Dear **Subhash Chandra Yadav**,

Following our selection process, we are pleased to offer you an internship position at **Thiranex**. We believe your skills and enthusiasm will be a great addition to our team.

|                   |                          |
|-------------------|--------------------------|
| Internship Role   | Intern - Web Development |
| Commencement Date | 30 Jul 2026              |
| Completion Date   | 29 Aug 2026              |
| Work Mode         | Remote / Project-Based   |

During this tenure, you will work on practical projects under industry mentorship. Your progress will be reviewed periodically, and your performance will determine the successful completion of the internship.

We look forward to a mutually beneficial learning journey. Please confirm your acceptance by starting your onboarding tasks.



This is an electronically generated document. No physical signature is required.  
For verification, contact Thiranex Verification Cell.  
www.thiranex.in



### Annexure I: Internship Offer Letter



**Annexure II: Certificate of Internship Completion**

## 4. Project Description

### 4.1 Introduction

This report documents the work undertaken during a Web Development Internship, completed with Thiranex – Skill Development & Future Tech, sourced via Internshala, as part of the academic requirement for the 2nd semester B.Tech CSE curriculum at IILM University, Greater Noida. The internship combined structured, milestone-based assignments with hands-on, project-based learning, and ran from 30 July 2026 to 29 August 2026 in Remote / Project-Based mode (Intern ID: THX-JUL3026-504).

As part of the practical component, five progressive modules were completed, each building directly on the skills developed in the previous one: (i) semantic and accessible HTML5 structure, (ii) advanced, responsive CSS3 styling, (iii) interactive JavaScript logic and client-side state management, (iv) asynchronous JavaScript and RESTful API integration, and (v) full-stack deployment and project architecture. Together, these modules trace the complete lifecycle of building and shipping a modern web application, culminating in a live, publicly deployed capstone project. Each module is described in detail in Section 4.8 (Methodology), and a module-wise summary of the internship curriculum is presented in Section 4.10 (Internship Training Log Book).

### 4.2 Organization Profile

Thiranex – Skill Development & Future Tech is a skill-development and future-technology focused platform that offers structured, project-based internship programs in modern web development. The organization's programs are designed to build industry-relevant, hands-on competencies through progressively advanced, real-world coding assignments, with continuous progress tracking through an online milestone-submission portal. The internship undertaken for this report was carried out entirely in Remote / Project-Based mode, under an arrangement facilitated via Internshala, and ran from 30 July 2026 to 29 August 2026.

### 4.3 Problem Statement

The internship's practical component was centred on building a complete, personal portfolio-style web application, one module at a time, with each stage addressing a distinct engineering problem:

- **Accessible Structure:** Given the requirement for a multi-page personal portfolio, the task was to build a semantically correct, WCAG-accessible HTML5 skeleton that would score well on automated accessibility and SEO audits.
- **Responsive Styling:** Given the semantic skeleton, the task was to design a fully responsive, theme-aware visual layer using advanced CSS3 (Grid, Flexbox, custom properties) that adapts across mobile, tablet and desktop viewports.
- **Client-side Interactivity:** Given the need for a stateful, data-driven interface, the task was to build a To-Do List application implementing full CRUD functionality with persistent local storage.
- **Live Data Integration:** Given a public REST API, the task was to build a Weather Dashboard capable of asynchronously fetching, parsing and rendering real-time JSON data, with robust error handling.
- **Production Deployment:** Given all previously built components, the task was to architect a modular, capstone E-commerce product catalog application and deploy it live to a modern hosting platform.

Together, these problems are representative of the standard front-end web development lifecycle: markup, styling, interactivity, data integration, and deployment.

#### 4.4 Project Objectives

- To understand and apply the complete front-end web development workflow, from semantic markup to a live, deployed application.
- To build strong foundations in semantic HTML5 and accessibility-first (WCAG/ARIA) markup.
- To design responsively using modern CSS3 (Grid, Flexbox, custom properties) for adaptive, theme-aware interfaces.
- To engineer interactive, stateful client-side applications using core JavaScript, the DOM, and the Web Storage API.
- To integrate live data by consuming real-world REST APIs asynchronously, with robust error handling.
- To gain practical, hands-on exposure to modern deployment platforms and production optimization practices.

#### 4.5 Scope of the Project

The scope of the internship is limited to an educational, portfolio-oriented implementation using client-side (front-end) technologies, and is not intended to represent a full production software



system with a dedicated back-end or database layer. The HTML5/CSS3 portfolio, the To-Do List application, and the Weather Dashboard are each scoped as independent, self-contained front-end projects. The final capstone project (Module 5) integrates the modular front-end architecture and deployment practices learned across all previous modules into a single, publicly accessible, production-ready web application. The focus throughout was on correctly applying front-end engineering fundamentals and deployment practices rather than on building a large-scale, multi-service back-end system.

## 4.6 Technologies and Tools Used

- Markup & Styling: HTML5 (semantic elements, ARIA), CSS3 (Grid, Flexbox, custom properties, media queries)
- Scripting: JavaScript (ES6+), DOM APIs, event delegation, async/await, the Fetch API
- Client-side Persistence: Browser Web Storage API (localStorage)
- Data Integration: Public REST APIs (JSON)
- Auditing: Lighthouse (Accessibility & SEO auditing)
- Deployment Platforms: Vercel / Netlify / Render
- Development Environment: Modern code editor with live-reload / browser developer tools

## 4.7 System Architecture

The overall internship followed the layered front-end architecture shown in Fig. 4.1: a semantic HTML5 structure forms the foundation, over which a responsive CSS3 presentation layer is applied; interactivity and state are then added using JavaScript; live data is integrated asynchronously via REST APIs; and the completed, integrated application is finally optimized and deployed to a production hosting platform.

|                                           |                                      |                                                       |                                                  |                                     |                                  |
|-------------------------------------------|--------------------------------------|-------------------------------------------------------|--------------------------------------------------|-------------------------------------|----------------------------------|
| <b>Semantic<br/>Structure<br/>(HTML5)</b> | <b>Responsive<br/>Styling (CSS3)</b> | <b>Client-side<br/>Interactivity<br/>(JavaScript)</b> | <b>Live Data<br/>Integration<br/>(REST APIs)</b> | <b>Optimization<br/>&amp; Build</b> | <b>Production<br/>Deployment</b> |
|-------------------------------------------|--------------------------------------|-------------------------------------------------------|--------------------------------------------------|-------------------------------------|----------------------------------|

*Fig. 4.1: Layered Front-End Architecture Followed Across the Internship*

## 4.8 Methodology

This section describes, module by module, the methodology followed to complete each of the five internship assignments.

### 4.8.1 Module 1: HTML5 Semantic Structure & Accessibility

**Due Date: 06 Aug 2026 | Status: Completed Successfully**

- Structuring page content using semantic HTML5 tags (header, nav, main, section, article, aside, footer) in place of generic <div> elements, to improve document structure and machine-readability.
- Applying ARIA labels and roles to interactive and non-standard elements to ensure compatibility with assistive technologies such as screen readers.
- Writing optimized, SEO-friendly meta tags (title, description, Open Graph tags) for each page to improve discoverability.
- Designing and implementing an accessible, keyboard/tab-navigable contact form with properly associated <label> elements.
- Validating the resulting markup and running Lighthouse Accessibility and SEO audits to confirm compliance.

*Outcome: Built a fully accessible, semantic multi-page portfolio skeleton scoring 100 on Lighthouse Accessibility and SEO audits.*

### 4.8.2 Module 2: Advanced CSS3 & Responsive Architecture

**Due Date: 13 Aug 2026 | Status: Completed Successfully**

- Converting the semantic HTML skeleton from Module 1 into a fully styled interface using CSS3.
- Applying CSS Grid to construct complex, two-dimensional page layouts (e.g., project galleries, multi-column sections).
- Using Flexbox for localized, one-dimensional component alignment (navigation bars, card layouts, form rows).
- Implementing mobile-first responsive design through progressively enhanced media queries.
- Defining a set of CSS custom properties (variables) to drive a dynamic light/dark mode theme switchable at runtime.

*Outcome: Delivered a pixel-perfect, fully responsive website that adapts seamlessly across mobile, tablet and desktop breakpoints.*

### 4.8.3 Module 3: JavaScript Logic & State Management

**Due Date: 20 Aug 2026 | Status: Completed Successfully**

- Designing the data model for a task (To-Do) item and the in-memory application state.

- Implementing full CRUD (Create, Read, Update, Delete) functionality using vanilla JavaScript and DOM APIs.
- Persisting application state automatically to the browser using the window.localStorage API, and rehydrating state on page load.
- Building advanced filtering (All / Active / Completed) driven by the underlying state rather than direct DOM queries.
- Using event delegation on a parent container to efficiently handle events for dynamically created list items.

*Outcome: Delivered a fully functional, state-driven To-Do List application that persists user data across browser reloads.*

#### **4.8.4 Module 4: Asynchronous JavaScript & RESTful APIs**

**Due Date: 27 Aug 2026 | Status: Completed Successfully**

- Studying the request/response cycle of RESTful APIs and the structure of the chosen public weather API.
- Implementing data fetching using the modern Fetch API in conjunction with async/await syntax for readable asynchronous code.
- Adding comprehensive error handling (try/catch, HTTP status checks) for failed or malformed network requests.
- Parsing and dynamically rendering complex, nested JSON response objects into the DOM.
- Implementing a search feature allowing the user to retrieve weather data for a city of their choice.

*Outcome: Delivered a dynamic Weather Dashboard that displays accurate, live weather metrics based on user-entered city names.*

#### **4.8.5 Module 5: Full-Stack Deployment & Project Architecture**

**Due Date: 29 Aug 2026 | Status: Completed Successfully**

- Architecting a modular frontend application by decomposing the capstone project (an E-commerce product catalog) into reusable components/partial.
- Implementing client-side routing to enable seamless navigation between views without full page reloads.
- Optimizing static assets (image compression, code minification) to improve load performance.

- Configuring a production build and deploying the completed application to a modern hosting platform (Vercel / Netlify / Render).
- Performing a final end-to-end review of all five modules to ensure consistency, accessibility and responsiveness across the integrated project.

*Outcome: Delivered a live, production-ready E-commerce product catalog capstone project with a public URL.*

## 4.9 Expected Outcomes

- A fully accessible, semantic HTML5 portfolio skeleton scoring 100 on Lighthouse Accessibility and SEO audits.
- A pixel-perfect, fully responsive presentation layer adapting flawlessly across mobile, tablet and desktop screens, with a working light/dark theme.
- A fully functional, state-driven To-Do List application with complete CRUD functionality and persistent local storage.
- A dynamic Weather Dashboard displaying accurate, live weather metrics based on user-entered city names, with robust error handling.
- A live, production-ready E-commerce product catalog capstone project with a public URL, demonstrating professional-grade front-end development and deployment capability.
- Practical, transferable familiarity with the complete front-end web development workflow — semantic markup, responsive styling, client-side scripting, API integration, and deployment — using industry-standard tools.
- A foundation for further exploration of back-end technologies, frameworks (e.g., React), and larger, real-world full-stack projects in future coursework.

## 4.10 Challenges Faced & Key Learnings

A number of practical challenges were encountered and resolved over the course of the internship; these are summarised below along with the corresponding learning outcome.

### 4.10.1 Debugging Asynchronous Logic

While building the Weather Dashboard (Module 4), race conditions and silently failing network requests made debugging difficult in the early stages. This was resolved by consistently wrapping asynchronous calls in try/catch blocks and explicitly checking HTTP response status codes before parsing JSON, which led to a much clearer understanding of how to trace and handle asynchronous failures gracefully using async/await.

### ***4.10.2 Designing for All Screen Sizes***

Retrofitting responsiveness after building a desktop-first layout in earlier attempts led to inconsistent results. Adopting a strict mobile-first workflow — starting from the smallest breakpoint and progressively enhancing the layout via media queries — proved far more reliable, and this iterative, breakpoint-first approach was carried through the remainder of the internship.

### ***4.10.3 Managing Application State***

In the initial implementation of the To-Do List application (Module 3), direct, ad-hoc DOM updates scattered throughout the code made the application difficult to reason about and prone to state/UI mismatches. Refactoring towards a single in-memory state object, with the DOM re-rendered from that state on every change, produced a much more predictable and maintainable state-and-render pattern.

### ***4.10.4 Shipping to Production***

Deploying the capstone project (Module 5) for the first time surfaced practical issues around asset paths, environment configuration, and build optimization that do not arise during local development. Working through these issues provided first-hand experience with production build pipelines, minification, and the deployment workflow of modern hosting platforms such as Vercel, Netlify and Render.

## **4.11 Skills & Competencies Gained**

### ***4.11.1 Front-End Development***

- Semantic HTML5 and ARIA-based accessibility practices.
- Responsive design using CSS Grid, Flexbox and mobile-first media queries.
- CSS custom properties and dynamic theming (light/dark mode).

### ***4.11.2 JavaScript Engineering***

- DOM manipulation and delegated event handling.
- Client-side state management and local data persistence.
- Asynchronous programming with `async/await`, the Fetch API, and error handling.

### ***4.11.3 Professional Practices***

- Modular, maintainable project architecture.
- Client-side routing and asset/performance optimization.
- End-to-end deployment of a live, production-ready web application.

## 4.12 Internship Training Log Book

The internship curriculum was delivered through five structured, milestone-based modules, each with a defined due date, a set of key topics/features, and an expected outcome. Table 4.1 summarises the module-wise progress.

| Module | Module Title                                            | Due Date    | Key Topics Covered                                                                                                                                                           | Key Learning Outcome                                                                                                      |
|--------|---------------------------------------------------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 1      | <b>HTML5 Semantic Structure &amp; Accessibility</b>     | 06 Aug 2026 | Semantic HTML5 elements (header, nav, main, section, article, footer); WCAG accessibility guidelines; ARIA roles and labels; SEO-friendly meta tags; accessible form design. | Built a fully accessible, semantic multi-page portfolio skeleton scoring 100 on Lighthouse Accessibility and SEO audits.  |
| 2      | <b>Advanced CSS3 &amp; Responsive Architecture</b>      | 13 Aug 2026 | CSS Grid layout; Flexbox; mobile-first media queries; CSS custom properties (variables); light/dark theming.                                                                 | Delivered a pixel-perfect, fully responsive website that adapts seamlessly across mobile, tablet and desktop breakpoints. |
| 3      | <b>JavaScript Logic &amp; State Management</b>          | 20 Aug 2026 | DOM manipulation; event delegation; client-side CRUD operations; the Web Storage API (localStorage); UI state and filtering logic.                                           | Delivered a fully functional, state-driven To-Do List application that persists user data across browser reloads.         |
| 4      | <b>Asynchronous JavaScript &amp; RESTful APIs</b>       | 27 Aug 2026 | The Fetch API; async/await syntax; error handling for network requests; JSON parsing; consuming public REST APIs.                                                            | Delivered a dynamic Weather Dashboard that displays accurate, live weather metrics based on user-entered city names.      |
| 5      | <b>Full-Stack Deployment &amp; Project Architecture</b> | 29 Aug 2026 | Modular frontend architecture; client-side routing; asset and code optimization; deployment platforms (Vercel / Netlify / Render).                                           | Delivered a live, production-ready E-commerce product catalog capstone project with a public URL.                         |

*Table 4.1: Internship Training Log Book – Module-wise Summary*

### 4.13 Certificates of Completion (Annexure at a Glance)

The Internship Offer Letter and the Certificate of Internship Completion issued by Thiranex – Skill Development & Future Tech are attached in full in the preceding section (Internship Completion Certificate) of this report as Annexure I and Annexure II respectively. A compact, at-a-glance view of both documents is provided below for quick reference.

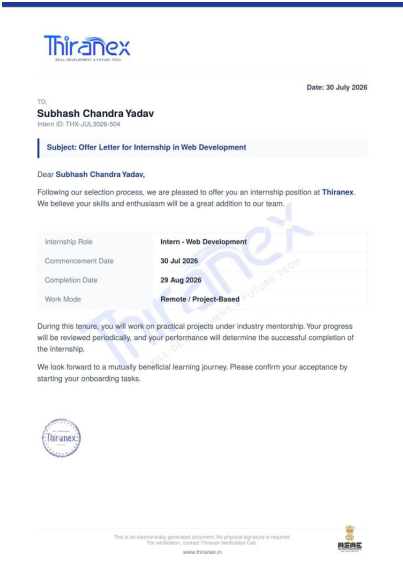
|                                                                                    |                                                                                    |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |
| <p><b>Annexure I: Internship Offer Letter</b></p>                                  | <p><b>Annexure II: Certificate of Completion</b></p>                               |

Table 4.2: Internship Offer Letter and Completion Certificate – Annexure Summary

## 5. Bibliography / References

- [1] Mozilla Developer Network (MDN), “HTML: HyperText Markup Language,” <https://developer.mozilla.org/en-US/docs/Web/HTML>
- [2] Mozilla Developer Network (MDN), “CSS: Cascading Style Sheets,” <https://developer.mozilla.org/en-US/docs/Web/CSS>
- [3] Mozilla Developer Network (MDN), “JavaScript Guide,” <https://developer.mozilla.org/en-US/docs/Web/JavaScript/Guide>
- [4] W3C, “Web Content Accessibility Guidelines (WCAG) 2.1,” <https://www.w3.org/WAI/standards-guidelines/wcag/>
- [5] Google Developers, “Lighthouse: Automated Auditing for Web Pages,” <https://developer.chrome.com/docs/lighthouse/overview/>
- [6] MDN Web Docs, “Using the Fetch API,” [https://developer.mozilla.org/en-US/docs/Web/API/Fetch\\_API/Using\\_Fetch](https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API/Using_Fetch)
- [7] MDN Web Docs, “Window: localStorage property,” <https://developer.mozilla.org/en-US/docs/Web/API/Window/localStorage>
- [8] Vercel Inc., “Vercel Documentation,” <https://vercel.com/docs>
- [9] Thiranex – Skill Development & Future Tech, Internship Program Materials, Task Briefs and Milestone Portal, Web Development Virtual Internship, 2026.